Using the ANOVA Test to Analyze Weekly Work Hours Based on Education Level in the USA (1994)

Agon Selmani

2026-11-05

Introduction

In this project, we analyze the weekly work hours in the United States for the year 1994, using the ANOVA test. This test, which stands for Analysis of Variance, is used to compare the means between three or more groups simultaneously and to determine if the differences between them are statistically significant. The objective is to understand whether there are significant differences in weekly work hours among individuals with different education levels.

Data Description

The data for this project includes socio-economic data from the 1994 US Census. The dataset is taken from Kaggle at the following link: Adult Income Dataset.

The analysis includes 32,561 observations (individuals). From this dataset, two main variables were selected:

- **Dependent Variable (continuous):** `hours.per.week` (Weekly work hours).
- **Independent Variable (categorical):** `education` (Attained education level).

Because the `education` variable had 16 different categories, they were grouped into 3 main categories to increase the accuracy of the statistical test: **High School**, **College**, and **Postgraduate**. Additionally, rows with missing values (denoted as "?") were removed to ensure the cleanliness of the model.

Hypothesis Formulation

In testing the hypotheses for this project, the goal is to compare the means of weekly work hours across the three education levels to see if there are statistically significant differences among them.

- **Null Hypothesis (H0):** The means of weekly work hours for the education levels (High School, College, Postgraduate) have no significant differences among themselves.
- **Alternative Hypothesis (H1):** At least one of the means of weekly work hours for the education levels has a significant difference from the others.

R Code Description

Data Preparation

Initially, the dataset was loaded and cleaned. Missing values ("?") were replaced with NA. The `education` variable had 16 categories, which were grouped into 3 main categories: High School, College, and Postgraduate. This was done to avoid groups with too few observations. Then, rows with missing values were removed, and the new column was converted to a factor.

```
# 1. Load the dataset
df <- read.csv("adult.csv", stringsAsFactors = FALSE)

# 2. Clean the data
# Replace "?" with NA
df[df == "?"] <- NA

# Remove rows with missing values (we only care about our variables for this test)
df <- na.omit(df[, c("education", "hours.per.week")])

# 3. Group education into broader categories (to make ANOVA reliable)
# We will create a new column called "edu_level"
df$edu_level <- "Other"
df$edu_level[df$education %in% c("HS-grad", "11th", "10th", "9th", "7th-8th", "12th", "1st-4th", "5th-6
df$edu_level[df$education %in% c("Some-college", "Assoc-voc", "Assoc-acdm")] <- "College"
df$edu_level[df$education %in% c("Bachelors", "Masters", "Prof-school", "Doctorate")] <- "Postgraduate"

# Convert to factor
df$edu_level <- as.factor(df$edu_level)
```

Unique values of the education variable

```
# Look at unique values of variable education
unique(df$education)
```

```
##  [1] "HS-grad"      "Some-college" "7th-8th"      "10th"         "Doctorate"
##  [6] "Prof-school"  "Bachelors"    "Masters"      "11th"         "Assoc-acdm"
## [11] "Assoc-voc"    "1st-4th"      "5th-6th"      "12th"         "9th"
## [16] "Preschool"
```

```
# See how many unique values it has
length(unique(df$education))
```

```
## [1] 16
```

Unique values of the edu_level variable

```
unique(df$edu_level)
```

```
## [1] High School  College      Postgraduate
## Levels: College High School Postgraduate
```

Descriptive Statistics

Before the ANOVA test, the mean and standard deviation were calculated for each group.

```
aggregate(hours.per.week ~ edu_level, data = df, FUN = mean)
```

```
##      edu_level hours.per.week
## 1      College       39.42464
## 2  High School       39.46340
## 3 Postgraduate       43.44180
```

```
aggregate(hours.per.week ~ edu_level, data = df, FUN = sd)
```

```
##      edu_level hours.per.week
## 1      College       12.48056
## 2  High School       12.07539
## 3 Postgraduate       12.19248
```

Data Visualization

A boxplot is used to compare the distribution of work hours for each education level, which is shown in the end.

ANOVA Test

The ANOVA analysis is performed using the `aov` function, where the means of work hours for different education levels are compared.

The test results are shown using the `summary` function.

```
anova_result <- aov(hours.per.week ~ edu_level, data = df)
summary(anova_result)
```

```
##                Df  Sum Sq Mean Sq F value Pr(>F)
## edu_level       2   96803   48402   323.8 <2e-16 ***
## Residuals   32558 4867262     149
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Post-hoc Test (Tukey HSD)

To identify exactly which groups have differences among them, Tukey's Post-hoc test was also performed.

```
# 7. Post-hoc Test (Tukey HSD)
TukeyHSD(anova_result)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = hours.per.week ~ edu_level, data = df)
##
## $edu_level
##                               diff        lwr       upr     p adj
## High School-College      0.0387591 -0.3353611 0.4128793 0.9680223
## Postgraduate-College     4.0171593  3.5857639 4.4485547 0.0000000
## Postgraduate-High School 3.9784002  3.5815998 4.3752005 0.0000000
```

Interpretation of Results

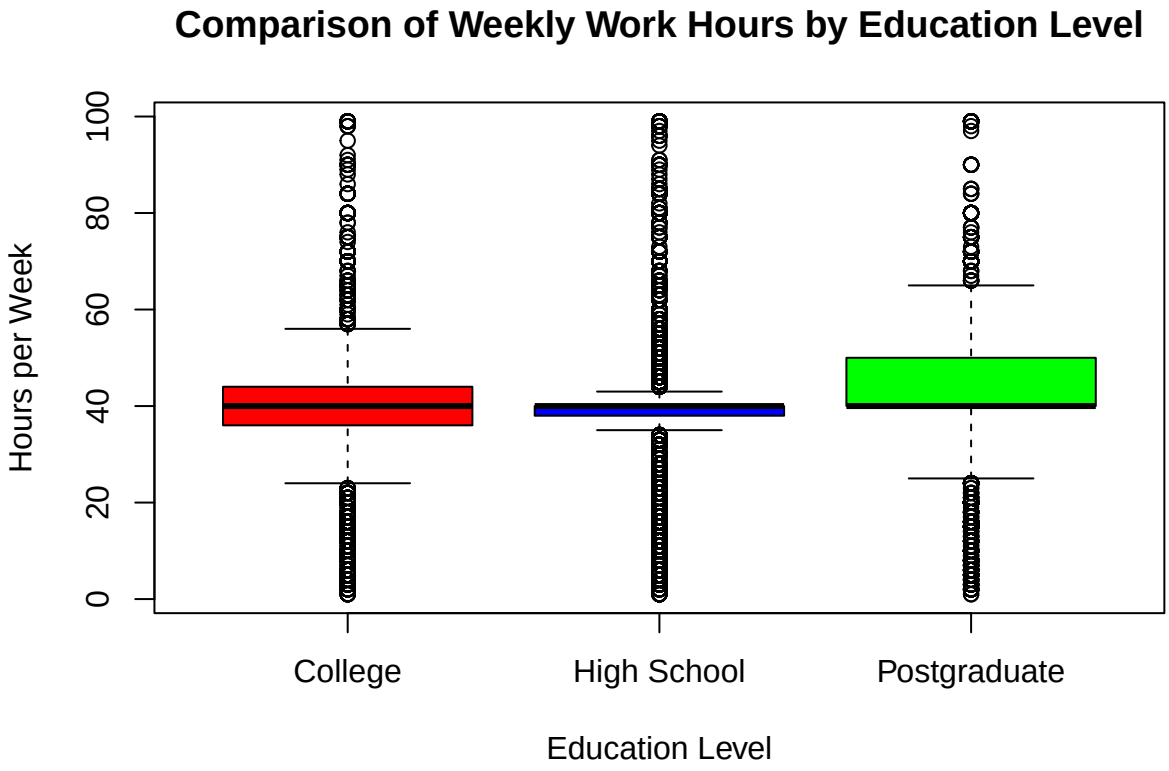
After implementing the ANOVA test, we can interpret the obtained results. From the test results, the p-value is $<$ 2e-16, which is much smaller than 0.05, indicating that we should reject the Null Hypothesis and accept the Alternative Hypothesis.

This means there are significant differences between the means of weekly work hours for different education levels.

From the Post-hoc test (Tukey HSD), we observe that the difference is concentrated in the group with postgraduate education (Postgraduate). They work on average about 4 hours more per week (43.44 hours) than individuals with higher education (College - 39.42 hours) or secondary education (High School - 39.46 hours). This difference is statistically significant ($p < 0.05$). Meanwhile, there is no significant difference between individuals with secondary education and those with higher education ($p = 0.968$).

The boxplot below shows the distribution of work hours for the three education levels. The green box (Postgraduate) is visibly shifted upwards, indicating a higher median. The red (College) and blue (High School) boxes are almost identical, confirming the results of the statistical test. The presence of outliers indicates that there are individuals working very few or very many hours, but this does not affect the accuracy of the test due to the large sample size.

This result reflects the economic theories of human capital, where individuals with higher qualifications tend to be employed in managerial or professional positions that require longer time commitments. However, it should be noted that this is an associative study and not causal; we cannot say with certainty that higher education *causes* longer work hours, but only that there is a strong statistical link between them in the 1994 data.


Comparison of Weekly Work Hours by Education Level
Hours per Week
0
20
40
60
80
100
College
High School
Postgraduate
Education Level